Week 7 Report: Neural Network Variations and Evaluation

1. Modeling Approach

This week we built a **multilayer perceptron (MLP) neural network** and evaluated three variations of it on the same 27-feature reduced set prepared in Week 5. We chose this approach for four reasons:

- **A natural step up in complexity.** Week 6 compared a linear model with two tree ensembles. A neural network is the next level of flexibility, and it tests whether a more complex model finds signal that logistic regression and the tree models missed.
- **It can learn interactions on its own.** Week 3 showed that response is not additive: single customers with no children respond far more than either factor alone suggests, and in Week 5 we had to build Lives_Alone by hand to capture it. A hidden layer can learn combinations like this without being told.
- **It produces probabilities that can be ranked.** This matches the business use, where the model's job is to put the most promising customers first.
- **The data is already in the right form.** Neural networks are sensitive to feature scale. Weeks 4 and 5 log-transformed the skewed columns and standardized the features, and the pipeline standardizes again inside each training fold so the validation fold never influences the scaling.

We scored the same two reference points as in Week 6: logistic regression with default settings (the model used in Weeks 4 and 5 to compare data choices) and the prior-campaign rule, which ranks customers by how many earlier campaigns they accepted. We also compare the results with the tuned logistic regression from Week 6 (validation PR-AUC 0.732), the best model so far.

Validation design. As decided in Week 3, we did not carve out a separate validation set, because it would hold only about 47 responders and give unstable estimates. Instead, each model was trained on four-fifths of the training set and scored on the remaining fifth (the validation fold), using 5-fold stratified cross-validation repeated 3 times, for 15 train/validation splits. Every model used the same 15 splits as Week 6, so results can be compared fold by fold. Training metrics are measured on the data each model learned from; validation metrics are measured on customers the model did not see. The test set was not used this week and is kept for the final evaluation.

Should we use a sequence model? Our professor suggested a model for sequential data, which would read the five earlier campaign columns in order and use the pattern in which a customer responded as a predictor. We assessed this idea against our data before choosing the model, and decided against it for four reasons:

- **The order is not confirmed.** The columns are numbered AcceptedCmp1 to AcceptedCmp5, but the dataset has no campaign dates, so we cannot verify that they are in time order.
- **A zero is ambiguous.** It can mean the customer declined, or that they were never contacted or had not yet become a customer. A pattern of zeros and ones is therefore partly a pattern of who was contacted.

- **There is very little sequence to learn from.** Each customer has only five 0/1 values, so just 32 histories are possible, and 1,270 of the 1,608 training customers (79.0%) have all zeros. Only 338 customers accepted at least one earlier campaign.
- **Most of the signal is in the count.** Response rises steadily with the number of campaigns accepted, from 7.6% with none to 30.9% with one, 50.0% with two, and 76.5% with three. Week 5 also showed that replacing the five flags with their total did not hurt performance.

We also tested the idea directly. Among the 233 customers who accepted exactly one earlier campaign, response did not rise with the campaign number: it was 26.3% for campaign 1, 42.4% for campaign 3, 12.9% for campaign 4, and 35.0% for campaign 5 (only one customer accepted campaign 2 alone). If later campaigns were more informative, the rate would climb toward campaign 5. The differences more likely reflect what each campaign offered than the order in which they ran.

We then added three order features to a logistic regression on the same 15 folds: the most recent campaign accepted, acceptances in the last two campaigns, and acceptances in the first two. Validation PR-AUC was 0.730, against 0.729 with the count alone. That is a gain of 0.001, better on only 9 of 15 folds, and using the five flags separately scored lower (0.725). We therefore kept the 27-feature set and used a standard neural network instead of a sequence model. The supporting tables are in Appendix A.

2. Model Complexity

A neural network is the most complex approach we have used so far. Logistic regression learns one coefficient per feature plus an intercept, 28 parameters in total. A neural network learns a weight for every connection between layers, so even a small network has hundreds of weights. The three variations differ sharply:

- **Variation 1** has one hidden layer of 8 units and learns 233 weights, about one per responder in the training set (234).
- **Variation 2** has one hidden layer of 16 units and learns 465 weights, about two per responder.
- **Variation 3** has two hidden layers of 64 and 32 units and learns 3,905 weights. That is about 17 per responder and more than twice the number of training customers (1,608).

The added complexity has three effects. First, the network is non-linear, so it can learn interactions between features that logistic regression cannot unless they are built in as features. Second, it is trained by repeatedly adjusting its weights from a random starting point, so results can vary slightly between runs; we fixed the random seed (42) so they can be reproduced. Third, it is the hardest model so far to explain, because no single coefficient shows how a feature raises or lowers the chance of response.

Complexity matters here because the data is small. With 234 responders, a flexible model has few examples per pattern it tries to learn, so it is more likely to fit noise. Variation 3 shows this clearly: it scored a training PR-AUC of 1.000, which means it memorized the training customers (Section 6).

3. Hyperparameters Evaluated

Unlike Week 6, which searched a grid of settings, this week we compared three deliberately spaced variations to show how complexity affects overfitting. Each variation changes two hyperparameters, the size of the network and the strength of its regularization:

Hyperparameter	Variation 1	Variation 2	Variation 3	What it controls and why we varied it
Hidden layer sizes	8	16	64, 32	Network size sets how much the model can memorize. Larger networks can learn more interactions but are more likely to fit noise with only 234 responders.
Alpha (L2 penalty)	3.0	1.0	0.001	Penalizes large weights, which forces a simpler, smoother model. Higher values protect against overfitting; very low values leave the network free to memorize.
Weights learned	233	465	3,905	The result of the two settings above, shown to compare complexity.

We varied these two because they are the settings that most directly control overfitting in a small-data neural network, and they move together as a complexity dial. Variation 1 is small and strongly regularized, so it is closest to logistic regression in what it can learn. Variation 3 is large and lightly regularized, so it is the most flexible and the most likely to memorize. Variation 2 sits in between.

All other settings were held fixed so the comparison isolates complexity: ReLU activation, the Adam optimizer, a learning rate of 0.001, up to 500 training passes, a random seed of 42, and a standard scaler fit inside each training fold. We did not use class weights or SMOTE. Week 5 showed that neither improved PR-AUC, the network has no class-weight option, and PR-AUC depends on how customers are ranked rather than on the class balance.

4. Performance Metrics

Only 14.6% of training customers responded. A model that predicts "no response" for everyone would be 85.4% accurate while finding no responders, so accuracy is misleading here. We used the same five metrics as Week 6, so the neural networks can be compared directly with the earlier models:

- **PR-AUC (average precision), our main metric.** It measures how well the model ranks responders above non-responders, with emphasis on the minority class. Random guessing scores 0.146, the response rate. This matches the business use: the model's job is to rank customers so the

company contacts the most promising ones first. It also suits a neural network, which outputs a probability for every customer.

- **ROC-AUC.** The probability that a randomly chosen responder is ranked above a randomly chosen non-responder. It is widely reported, but with imbalanced classes it can look high even when precision is poor, so we treat it as secondary.
- **Precision.** Of the customers the model flags, the share who actually respond. Low precision means wasted contact costs.
- **Recall.** Of all responders, the share the model flags. Low recall means missed revenue.
- **F1 score.** The harmonic mean of precision and recall, summarizing the trade-off between them in one number.

PR-AUC and ROC-AUC use the predicted probabilities directly. Precision, recall, and F1 require a cutoff to turn probabilities into contact decisions, and in the comparison below they use a cutoff of 0.5. Because the three networks produce probabilities with different spreads, a fixed cutoff does not treat them equally, which is why we rely on PR-AUC for the main comparison.

5. Results: Training and Validation Metrics

Variation 1 scored highest on validation PR-AUC (0.725), just ahead of Variation 2 (0.722) and well ahead of Variation 3 (0.678). All values are averages over the 15 cross-validation folds; precision, recall, and F1 use a 0.5 cutoff. Bold marks the best validation value for each metric among the three variations.

Model	Dataset	PR-AUC	ROC-AUC	Precision	Recall	F1
Variation 1 (8 units, alpha 3.0)	Training	0.779	0.943	0.832	0.527	0.645
Variation 1 (8 units, alpha 3.0)	Validation	**0.725**	**0.928**	**0.780**	0.486	0.596
Variation 2 (16 units, alpha 1.0)	Training	0.867	0.968	0.867	0.654	0.745
Variation 2 (16 units, alpha 1.0)	Validation	0.722	0.927	0.750	0.551	**0.632**
Variation 3 (64 and 32 units, alpha 0.001)	Training	1.000	1.000	1.000	1.000	1.000
Variation 3 (64 and 32 units, alpha 0.001)	Validation	0.678	0.914	0.673	**0.585**	0.623

Reference points (same 15 folds):

Reference	Dataset	PR-AUC	ROC-AUC	Precision	Recall	F1
Logistic regression (default settings)	Training	0.769	0.941	0.789	0.565	0.658
Logistic regression (default settings)	Validation	0.729	0.930	0.743	0.547	0.627
Logistic regression (tuned, Week 6)	Validation	0.732	0.930	0.740	0.558	0.633
Prior-campaign rule	Validation	0.395	-	-	-	-
Random guessing	Validation	0.146	0.500	-	-	-

6. How Training and Validation Metrics Change Across Variations

The more complex the network, the higher its training score and the lower its validation score. The gap between training and validation PR-AUC shows how much each variation overfits:

Model	Training PR-AUC	Validation PR-AUC	Gap	Validation SD across folds
Variation 1 (8 units, alpha 3.0)	0.779	0.725	0.054	0.061
Variation 2 (16 units, alpha 1.0)	0.867	0.722	0.145	0.053
Variation 3 (64 and 32 units, alpha 0.001)	1.000	0.678	0.322	0.059
Logistic regression (default settings)	0.769	0.729	0.040	0.070

- **Training scores rise with complexity.** Training PR-AUC goes from 0.779 to 0.867 to 1.000. Variation 3 scored 1.000 on every training metric, which means it classified every training customer correctly. It memorized the training data rather than learning patterns that carry over.
- **Validation scores fall.** Validation PR-AUC goes from 0.725 to 0.722 to 0.678, and ROC-AUC from 0.928 to 0.927 to 0.914. Moving from Variation 1 to 2 raised training PR-AUC by 0.088 but lowered validation PR-AUC by 0.003, so the extra capacity mostly fit noise. Moving to Variation 3 cost a further 0.044 on validation.
- **The gap widens from 0.054 to 0.145 to 0.322.** Variation 3's gap is as large as the random forest's in Week 6 (0.318), and Variation 1's gap is close to logistic regression's (0.040). A small, strongly regularized network behaves almost like logistic regression because it has little room to learn more.
- **Precision and recall trade off at the 0.5 cutoff.** Validation precision falls from 0.780 to 0.750 to 0.673 while recall rises from 0.486 to 0.551 to 0.585. More complex networks produce more extreme probabilities, so more customers cross 0.5. Validation F1 is highest for Variation 2 (0.632), but F1 depends on the cutoff, whereas PR-AUC does not.

- **Fold-to-fold variation is larger than some of the differences.** Validation PR-AUC varies by about 0.05 to 0.06 across folds, far more than the 0.003 between Variations 1 and 2, so the difference between Variations 1 and 2 is within noise. Variation 3 is the only one that is clearly worse: it beat the default logistic regression on just 3 of 15 folds.

In short, extra complexity bought memorization, not signal. This repeats the Week 6 finding, where the tree ensembles fit the training data far better but generalized worse.

7. Validation Comparison and Best Model

On the validation folds, Variation 1 scored highest on PR-AUC (0.725), ROC-AUC (0.928), and precision (0.780). Variation 2 was second on PR-AUC (0.722) and had the highest F1 (0.632). Variation 3 was last on PR-AUC (0.678), although it had the highest recall (0.585) because its probabilities are the most extreme. Compared fold by fold, Variation 1 beat Variation 2 on 7 of 15 folds, with a mean difference of +0.003 and a fold-to-fold standard deviation of the difference of 0.022, so the two are effectively tied on score.

Best model: Variation 1 (one hidden layer of 8 units, alpha = 3.0). We selected it for three reasons:

1. **Best validation performance** on the main metric (PR-AUC 0.725), and on ROC-AUC and precision.
2. **Least overfitting.** Its train-validation gap of 0.054 is far smaller than Variation 2's (0.145) and Variation 3's (0.322), so its performance is the most likely to hold for new customers.
3. **Simplest model.** It learns 233 weights, about one per responder, while Variation 3's 3,905 weights memorized the training data.

Because its lead over Variation 2 is within the fold-to-fold noise, the choice rests as much on its lower overfitting and simplicity as on its score.

Comparison with Week 6. Variation 1 is the best neural network, but it is not better than logistic regression. Its validation PR-AUC of 0.725 is below both the default logistic regression (0.729) and the tuned version (0.732), and it beat the default logistic regression on only 6 of 15 folds (Variation 2 on 8 and Variation 3 on 3). So the extra complexity of a neural network did not add value on this dataset. This is likely because the main interactions are already captured by the Week 5 features (Income_per_Person and Lives_Alone), and with only 234 responders there is little additional signal for a flexible model to find. All three variations still beat the prior-campaign rule (0.395) by a wide margin, Variation 1 by 0.330, so they add real value over a simple business rule.

8. Conclusion

Variation 1, the smallest and most strongly regularized neural network, is the best model this week. It had the highest validation PR-AUC (0.725), the smallest gap between training and validation scores among the three (0.054), and the fewest weights. As the networks grew larger and less regularized, training scores rose to a perfect 1.000 while validation PR-AUC fell to 0.678, the same overfitting pattern we saw with the tree ensembles in Week 6.

The neural network did not beat logistic regression, which scored 0.732 when tuned and 0.729 with default settings, so logistic regression remains the strongest and easiest to explain model so far. Our check of the campaign history also gave no support for a sequence model: order features added only 0.001 to PR-AUC, and the history is too short and too ambiguous to learn from. The test set was not used this week and remains reserved for the final evaluation.

Appendix A: Campaign History and Sequence Check

Table A1 shows how many training customers accepted each earlier campaign, and how many of them responded to the last campaign. Table A2 groups customers by how many earlier campaigns they accepted.

Table A1. Acceptance by earlier campaign (1,608 training customers)

Campaign	Customers who accepted	Acceptance rate	Response rate among them
Campaign 1	113	7.0%	55.8%
Campaign 2	21	1.3%	76.2%
Campaign 3	117	7.3%	48.7%
Campaign 4	124	7.7%	35.5%
Campaign 5	120	7.5%	53.3%
Last campaign (Response)	234	14.6%	-

Table A2. Response by number of earlier campaigns accepted

Campaigns accepted	Customers	Share of customers	Response rate
0	1,270	79.0%	7.6%
1	233	14.5%	30.9%
2	62	3.9%	50.0%
3	34	2.1%	76.5%
4	9	0.6%	88.9%

Table A3 tests whether order matters beyond the count, using only the 233 customers who accepted exactly one earlier campaign. If later campaigns were more informative, the response rate would rise toward campaign 5. It does not.

Table A3. Customers who accepted exactly one earlier campaign

Campaign accepted	Customers	Response rate
Campaign 1	38	26.3%
Campaign 2	1	100.0%
Campaign 3	92	42.4%
Campaign 4	62	12.9%
Campaign 5	40	35.0%

Table A4 adds order features to a logistic regression and scores it on the same 15 folds as the other models.

Table A4. Do order features improve validation PR-AUC?

Features	PR-AUC	SD across folds	Change vs count only	Folds improved (of 15)
Count of campaigns accepted only (27 features)	0.729	0.067	-	-
Plus three order features (30 features)	0.730	0.068	+0.001	9
All five campaign flags separately (32 features)	0.725	0.066	-0.004	4

The three order features were the most recent campaign accepted, acceptances in campaigns 4 and 5, and acceptances in campaigns 1 and 2. We set the rule in advance: keep them only if they raised PR-AUC by at least 0.005 and improved at least 10 of 15 folds. They did neither, so the 27-feature set was used.

Appendix B: Figures

Figure B1. Training vs validation PR-AUC by model. The dashed line is the prior-campaign rule (0.395). Variation 3's training bar reaches 1.0, while its validation bar is the lowest of the three.

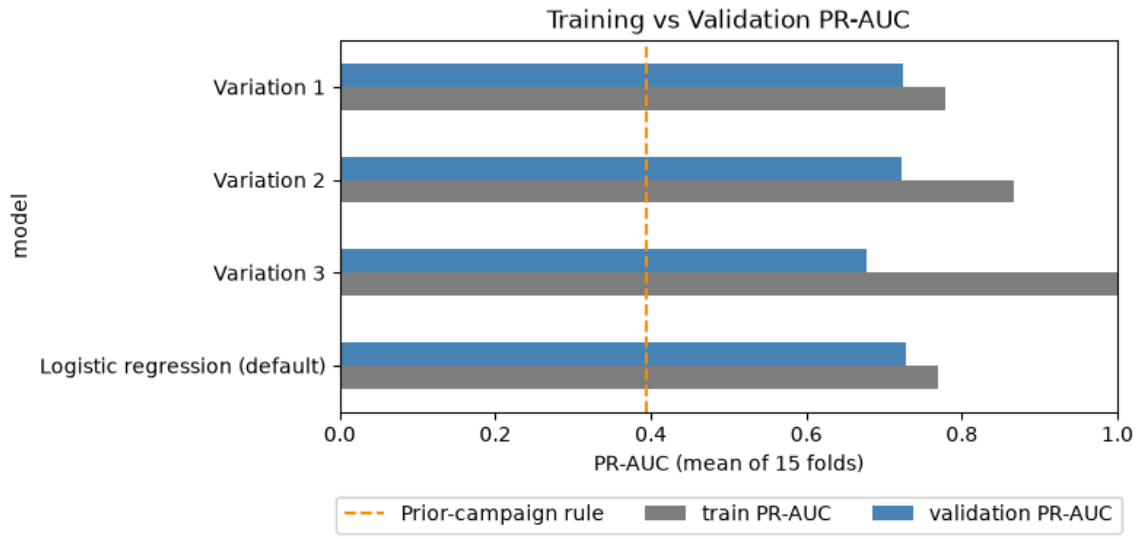


Figure B2. Precision-recall curves (left) and training loss (right). The curves use one pass of 5-fold cross-validated predictions, so their PR-AUC values (Variation 1 0.721, Variation 2 0.711, Variation 3 0.677, default logistic regression 0.724) differ slightly from the 15-fold averages in the tables. Variations 1 and 2 and the logistic regression nearly overlap, while Variation 3 sits lower across most of the recall range. On the right, Variation 3's training loss falls close to zero, which is another sign that it memorized the training customers. All three networks stopped on their own before the 500-pass limit.

